

Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	783
Is this Group for Examiner? :	No
Examiner permission :	Cant View
Show Progress Bar? :	No
Revisit allowed for group Instructions? :	Yes
Maximum Instruction Time :	0
Minimum Instruction Time :	0
Group Time In :	Minutes
Revisit Section :	Yes
Action on Revisit Section :	View and Edit
Navigate To Group Summary From Last Question? :	No
Disable Submit Button During Assessment? :	No
Section Selection Time? :	0
No of Optional sections to be attempted :	0

Sw Engg

Section Id :	640653121906
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	25
Number of Questions to be attempted :	25
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653317977
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406531815580 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SOFTWARE ENGINEERING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535902654. ✓ YES

6406535902655. ✗ NO

Sub-Section Number :

2

Sub-Section Id :

640653317978

Question Shuffling Allowed :

Yes

Question Number : 2 Question Id : 6406531815581 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A company is developing an *Online Food Delivery System (OFDS)*. Different users and systems interact with OFDS in various ways. Identify the *type of client* in each case.

1. A customer orders food from a restaurant using the OFDS mobile app.
2. The restaurant staff updates menu items and prices through the OFDS portal.
3. The payment gateway processes the customer's payment for the order.
4. The delivery tracking module communicates with the restaurant's order management system to update order status.

Which of the following correctly matches the *type of client* for each case?

Options :

6406535902656. ✓ 1 → External user, 2 → Internal user, 3 → Software client, 4 → Software client

6406535902657. ✗ 1 → Internal user, 2 → External user, 3 → Software client, 4 → Internal user

6406535902658. ✗ 1 → External user, 2 → Software client, 3 → Internal user, 4 → Software client

6406535902659. ✗ 1 → Software client, 2 → External user, 3 → Internal user, 4 → Internal user

Sub-Section Number :

3

Sub-Section Id :

640653317979

Question Shuffling Allowed :

Yes

Question Number : 3 Question Id : 6406531815582 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A ride-hailing company has completed the development of a fare estimation feature. Internal testers simulate trips and detect issues before release. Later, the company releases the feature to selected users to get real-world feedback.

Which testing phases do these correspond to?

Options :

6406535902660. ✖ Integration testing → Beta testing

6406535902661. ✔ Alpha testing → Beta testing

6406535902662. ✖ Unit testing → Integration testing

6406535902663. ✖ Beta testing → Alpha testing

Sub-Section Number :

4

Sub-Section Id :

640653317980

Question Shuffling Allowed :

Yes

Question Number : 4 Question Id : 6406531815583 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

In a distributed software development project, multiple teams work on different modules. Each module exposes an interface to allow other teams to interact with it.

From the following statements, which one(s) correctly describe the purpose of an interface definition?

Options :

6406535902664. ✔ It specifies what actions a function or module can perform.

6406535902665. ✖ It explains how the function or module is implemented internally.

6406535902666. ✔ It defines the allowed inputs, requests, and format of the outputs.

6406535902667. ✔ Once the interface is defined, the underlying implementation can change without affecting other modules.

Sub-Section Number :

5

Sub-Section Id :

640653317981

Question Shuffling Allowed :

Yes

Question Number : 5 Question Id : 6406531815584 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

A team is building a mobile banking app. The exact requirements are unclear, so they build a prototype with limited functionality and share it with the client for feedback before building the full system.

Which of the following are advantages of using a prototype model?

Options :

6406535902668. ✔ Helps clients understand what they actually need.

- 6406535902669. ✖ Reduces development cost because prototypes are always used in production.
- 6406535902670. ✔ Helps developers identify technical issues and improve design before full development.
- 6406535902671. ✖ Ensures all bugs are caught before the final system is built.

Sub-Section Number : 6
Sub-Section Id : 640653317982
Question Shuffling Allowed : Yes

Question Number : 6 Question Id : 6406531815585 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

You are developing features for the sales summary module of the seller portal. Different industries have different priorities: mobile sales teams care about launch day sales, whereas furniture sales teams care about 30-day sales trends.

Why might you conduct focus groups with sales managers from different industries?
(More than one answer may be correct)

Options :

- 6406535902672. ✔ To gain consensus on common requirements across industries.
- 6406535902673. ✔ To highlight conflicts or differences between stakeholder needs.
- 6406535902674. ✖ To observe how users perform tasks in their natural setting.
- 6406535902675. ✖ To get detailed regulatory information from banks.

Sub-Section Number : 7
Sub-Section Id : 640653317983
Question Shuffling Allowed : Yes

Question Number : 7 Question Id : 6406531815586 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following two requirements:

Requirement 1: The mobile banking app must complete a fund transfer transaction within 3 seconds.

Requirement 2: The mobile banking app must allow users to view their account balance and transaction history.

Which of the following statements is true?

Options :

- 6406535902676. ✖ Requirement 1 is a functional requirement, whereas Requirement 2 is a non-functional requirement.
- 6406535902677. ✔ Requirement 2 is a functional requirement, whereas Requirement 1 is a non-functional requirement.
- 6406535902678. ✖ Both requirements are functional requirements.

6406535902679. ✖ Both the requirements are non-functional requirements.

Sub-Section Number : 8
Sub-Section Id : 640653317984
Question Shuffling Allowed : Yes

Question Number : 8 Question Id : 6406531815587 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

Consider the user story given below for a **ride-sharing app** and answer the question.

Feature: Find nearby drivers

As a rider,
I want to search for available drivers
so that I can see driver details, estimated fare and arrival time within a reasonable time.
The results must be displayed with clear sorting and filtering options.

Now consider the following improvement in the user story. The phrase

so that I can see driver details, estimated fare and arrival time within a reasonable time.

is modified as:

so that I can see driver details, estimated fare and arrival time within 4 seconds.

It makes the user story more _____.

Options :

- 6406535902680. ✖ Specific
- 6406535902681. ✔ Measurable
- 6406535902682. ✖ Achievable
- 6406535902683. ✖ Relevant
- 6406535902684. ✖ Timeboxed

Sub-Section Number : 9
Sub-Section Id : 640653317985
Question Shuffling Allowed : Yes

Question Number : 9 Question Id : 6406531815588 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

A hospital management system introduces a new appointment booking interface. Users complain that it takes them a long time to figure out how to add or delete a patient record, and no tutorial or guided steps are provided.

Which usability goal is not satisfied?

Options :

- 6406535902685. ✓ Learnability
- 6406535902686. ✗ Memorability
- 6406535902687. ✗ Effectiveness
- 6406535902688. ✗ Efficiency

Sub-Section Number : 10
Sub-Section Id : 640653317986
Question Shuffling Allowed : Yes

Question Number : 10 Question Id : 6406531815589 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question
During a client meeting, the UX designer presents hand-drawn sketches of different screens on paper to demonstrate navigation flow. The client suggests some changes, which the designer applies immediately.

Which type of prototype is being used?

- Options :**
- 6406535902689. ✗ Storyboard
 - 6406535902690. ✓ Paper prototype
 - 6406535902691. ✗ Interactive prototype
 - 6406535902692. ✗ Digital mock-up

Sub-Section Number : 11
Sub-Section Id : 640653317987
Question Shuffling Allowed : Yes

Question Number : 11 Question Id : 6406531815590 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question
A new user opens a document editor and navigates menus to copy text, while an experienced user immediately uses Ctrl+C. Which design heuristic is being demonstrated?

- Options :**
- 6406535902693. ✗ Recognition over recall
 - 6406535902694. ✓ Flexibility
 - 6406535902695. ✗ Freedom
 - 6406535902696. ✗ Providing help

Sub-Section Number : 12
Sub-Section Id : 640653317988
Question Shuffling Allowed : Yes

Question Number : 12 Question Id : 6406531815591 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

A development team is creating a new e-commerce platform. They group related functionalities like inventory management, payment processing, and catalog management into separate modules. What aspect of the design process is being demonstrated?

Options :

- 6406535902697. ✖ Interfaces
- 6406535902698. ✔ Components
- 6406535902699. ✖ Algorithms
- 6406535902700. ✖ Data structures

Sub-Section Number : 13
Sub-Section Id : 640653317989
Question Shuffling Allowed : Yes

Question Number : 13 Question Id : 6406531815592 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

Module M1 directly modifies variables inside module M2 by referencing its internal implementation. What is this type of coupling?

Options :

- 6406535902701. ✖ Data coupling
- 6406535902702. ✖ Common coupling
- 6406535902703. ✔ Content coupling
- 6406535902704. ✖ Control coupling

Sub-Section Number : 14
Sub-Section Id : 640653317990
Question Shuffling Allowed : Yes

Question Number : 14 Question Id : 6406531815593 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

A developer changes code in one module, and multiple other modules start failing due to dependencies. Which issue does this illustrate?

Options :

- 6406535902705. ✖ Poor data cohesion
- 6406535902706. ✔ High coupling
- 6406535902707. ✖ High cohesion
- 6406535902708. ✖ Good modular design

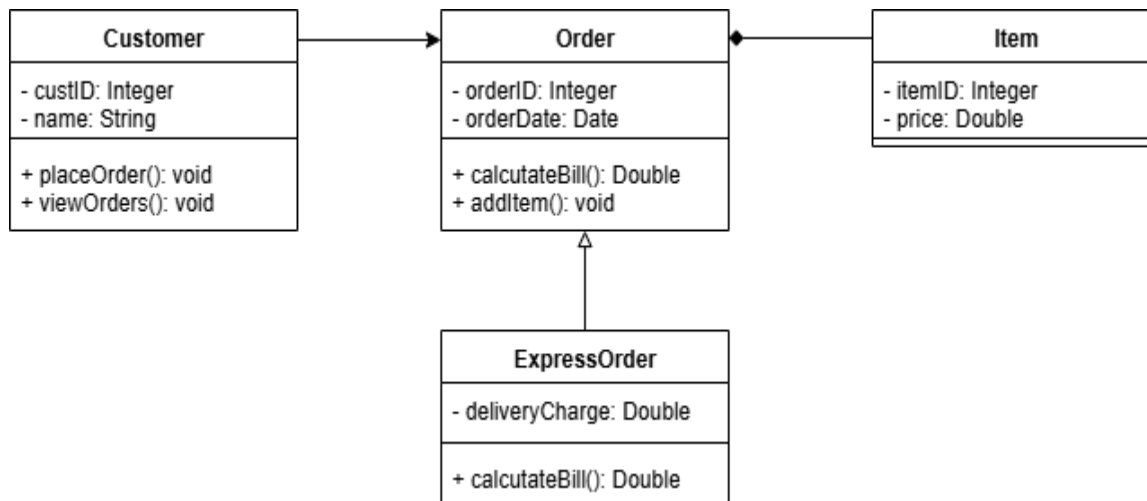
Sub-Section Number : 15
Sub-Section Id : 640653317991
Question Shuffling Allowed : Yes

Question Number : 15 Question Id : 6406531815594 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the class diagram given below.



Which of the following is/are true about the class diagram?

Options :

- 6406535902709. ✓ There is an association relation between Customer and Order.
- 6406535902710. ✓ There is a composition relation between Order and Item.
- 6406535902711. ✓ There is an inheritance relation between Order and ExpressOrder.
- 6406535902712. ✗ There is an aggregation relation between Customer and Order.
- 6406535902713. ✗ There is a composition relation between Customer and Order.
- 6406535902714. ✗ There is an inheritance relation between Item and Order.

Sub-Section Number :

16

Sub-Section Id :

640653317992

Question Shuffling Allowed :

Yes

Question Number : 16 Question Id : 6406531815595 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Which debugging strategy involves tracing the flow of execution from the beginning of the program to identify where the output deviates from expectations?

Options :

- 6406535902715. ✓ Forward debugging
- 6406535902716. ✗ Backward debugging
- 6406535902717. ✗ Input manipulation strategy
- 6406535902718. ✗ Black-box debugging

Sub-Section Number :

17

Sub-Section Id :

640653317993

Question Shuffling Allowed :

Yes

Question Number : 17 Question Id : 6406531815596 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A developer has written the following Python function to compute the factorial of a number:

```
def factorial(n):  
    if n < 0:  
        return None  
    elif n == 0 or n == 1:  
        return 1  
    else:  
        result = 1  
        for i in range(2, n + 1):  
            result *= i  
        return result
```

What is the cyclomatic complexity of this function?

Options :

6406535902719. ✖ 2

6406535902720. ✖ 3

6406535902721. ✔ 4

6406535902722. ✖ 5

Sub-Section Number :

18

Sub-Section Id :

640653317994

Question Shuffling Allowed :

Yes

Question Number : 18 Question Id : 6406531815597 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A developer is working on a branch `feature1` while another developer fixes a bug in branch `issueP` and merges it into the `master`. The developer now needs to use a function (`fun1`) added in the bug-fix branch, inside `feature1`.

Which Git command should be used before merging, to bring the latest changes from master into `feature1`?

Options :

6406535902723. ✖ `git fetch master`

6406535902724. ✖ `git pull origin feature1`

6406535902725. ✔ `git rebase master`

6406535902726. ✖ `git clone master`

Sub-Section Number :

19

Sub-Section Id :

640653317995

Question Shuffling Allowed :

Yes

Question Number : 19 Question Id : 6406531815598 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A compiler processes source code into assembly and then an executable file. Each step transforms data and passes it to the next stage. Which architectural style best describes this system?

Options :

6406535902727. ✖ Peer-to-Peer

6406535902728. ✖ Model-View-Controller

6406535902729. ✔ Pipe and Filter

6406535902730. ✖ Client-Server

Sub-Section Number :

20

Sub-Section Id :

640653317996

Question Shuffling Allowed :

Yes

Question Number : 20 Question Id : 6406531815599 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A developer is designing a software system for a marketplace. There are multiple delivery types: physical shipment, email delivery, and instant download.

Which of the following is a key advantage of applying the Interface Segregation Principle in this scenario?

Options :

6406535902731. ✖ Clients are forced to implement all delivery methods, ensuring consistency.

6406535902732. ✔ Each client implements only the methods relevant to its delivery type, avoiding meaningless implementations.

6406535902733. ✖ Interfaces are removed entirely, and all functionality is placed in one monolithic class.

6406535902734. ✖ It reduces the number of classes in the system.

Sub-Section Number :

21

Sub-Section Id :

640653317997

Question Shuffling Allowed :

Yes

Question Number : 21 Question Id : 6406531815600 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

You are designing a Book class with mandatory fields (ISBN, title, cost) and optional fields (author, published year, description). Using multiple constructors to handle every combination of fields leads to telescoping constructors.

Which design pattern would best solve this problem?

Options :

6406535902735. ✖ Factory Pattern

6406535902736. ✓ Builder Pattern

6406535902737. ✗ Adapter Pattern

6406535902738. ✗ Facade Pattern

Sub-Section Number : 22
Sub-Section Id : 640653317998
Question Shuffling Allowed : Yes

Question Number : 22 Question Id : 6406531815601 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

In a sprint planning meeting, which of the following are true?

Options :

6406535902739. ✓ The team decides what work can be completed in the sprint

6406535902740. ✗ Only the product owner decides which user stories are included

6406535902741. ✓ The team discusses how the selected work will be implemented

6406535902742. ✓ Sprint planning is time-boxed

Sub-Section Number : 23
Sub-Section Id : 640653317999
Question Shuffling Allowed : Yes

Question Number : 23 Question Id : 6406531815602 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a Work Breakdown Structure (WBS) for a project with root node "Amazon Seller Portal" and leaf nodes such as "Component Diagram," "Database Schema," and "User Interface." Which of the following statements are correct about WBS?

Options :

6406535902743. ✓ WBS breaks down project activities into smaller tasks

6406535902744. ✓ Each leaf node represents a task that can be assigned to a developer

6406535902745. ✗ Dependencies between tasks are explicitly represented in the WBS

6406535902746. ✓ WBS is structured like a tree with the project as the root

Sub-Section Number : 24
Sub-Section Id : 640653318000
Question Shuffling Allowed : Yes

Question Number : 24 Question Id : 6406531815603 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

A project manager has identified the following risks:

Risk	Probability (P)	Impact (I)
R1: Team member leaving	0.3	4
R2: Third-party module failure	0.5	3
R3: Ambiguous requirements	0.7	3

Which risk should be prioritized based on risk score?

Options :

6406535902747. ✖ R1

6406535902748. ✖ R2

6406535902749. ✔ R3

6406535902750. ✖ All equally

Sub-Section Number :

25

Sub-Section Id :

640653318001

Question Shuffling Allowed :

No

Question Id : 6406531815604 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (25 to 26)

Question Label : Comprehension

QWave Solutions is developing an E-commerce Seller Portal for a mid-sized online retailer. The system will include modules for catalog management, inventory tracking, order processing, payments, sales analytics, and customer feedback.

After analyzing past projects, the project manager estimates the total code size to be 6 KLOC. The project is classified as a semi-detached project, so the COCOMO parameters are:
 $a = 3.0$, $b = 1.12$.

After computing the initial effort, the team also identifies four important cost drivers:

Cost Driver	Rating	Multiplier
Required reliability	High	1.15
Database size	High	1.10
Team application experience	Low	1.12
Programming language experience	Nominal	1.00